Probabilistic Collaborative Representation Based Ensemble Learning for Classification of Wetland Hyperspectral Imagery

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***Abstract*— Protection of wetlands is important for ecosystem in recent years, and the classification of wetland ground cover is the foundation of investigation and protection work. Probabilistic collaborative representation classifier (ProCRC) is one of the best performing classifiers, which has been applied in the hyperspectral image (HSI) classification. However, its performance is greatly limited for wetland data where spectrums are highly similar. Moreover, the complex distribution of ground objects in wetlands has not been wisely utilized in the classification. In this article, the intrinsic mechanism of ProCRC is found and its kernel version is proposed to solve the problems of wetland classification. Then, a new ensemble learning strategy that considers neighborhood information is proposed, which largely alleviates the problem of sample collection in wetlands. Under the guidance of this strategy, two specific ensemble learning algorithms, i.e., local neighborhood-based ensemble (LNE) and local neighborhood and samples augmentation-based ensemble (LNSAE), are proposed. The superiority of the proposed methods is validated using three typical HSI datasets of China coastal wetland with few samples.**

***Index Terms*— Bagging, collaborative representation, ensemble learning, hyperspectral image (HSI) classification, kernel methods, wetlands.**

I. Introduction

WETLAND, as an important component of ecological environment, plays an important role in protecting biodiversity, regulating runoff, improving water quality, adjusting climate, and so on [1]. Recognition and classification of land surface types and species in wetlands is the premise of studying wetland ecology. Unlike other types of land surfaces, wetlands contain various ground objects with high similarity of spectrum [2], [3]. Compared with multispectral images (MSIs) using dozens of broadbands, hyperspectral remote sensing has been widely applied in various significant aspects such as environmental monitoring and accurate classification due to its availability of obtaining hundreds of narrow and continuous spectral bands [4], [5], [6], [7], [8], [9]. However, there are still some limitations in wetland classification tasks due to the complexity of wetland environment [10]. These challenges mainly come from the following aspects: small number and high acquisition cost of labeled samples, inseparability of data in the original band space, and inadequate utilization of unique spatial information existing in remote sensing images [11], [12].

Significant achievements have been made with machine learning methods, such as support vector machine (SVM), for wetland classification [13]. However, with the increase of data dimension, the accuracies of these methods tend to increase first and then decrease, which is called the "Hughes phenomenon [14]." Deep learning (DL) methods represented by convolutional neural networks (CNNs) are also increasingly applied to the classification of wetlands [15], [16], [17], [18]. Many parameters in a DL model need to be adjusted, which makes the training process very time-consuming. Also, it requires quantities of known samples, which are difficult to obtain in wetlands.

To provide a better solution for wetland classification, Su et al. [19] applied the low-rank representation (LRR) method to classify hyperspectral images (HSIs) of wetland and obtained promising results. This method is derived from the idea of compressed sensing, which claims that high-dimensional data mainly exists in low-dimensional subspaces [20]. It makes it possible to reconstruct testing samples by linear combinations of training samples. This type of method can alleviate the "curse of dimensionality" due to their compact models [21]. The representation models were successfully applied in HSI classification because of their strong robustness and convenience of avoiding complicated training process [22]. Sparse representation classification (SRC) [23] is one of the representation based methods, which has strong robustness but high computational complexity. Zhang et al. [24] argued that the success of SRC is mainly attributed to the collaborative representation of all training samples rather than the sparse representation of few samples. Collaborative representation classification (CRC) is then proposed [25]. Variants of CRC-based classifiers have subsequently emerged for HSI classification, for example, tangent CRC (TCRC) [26], kernel collaborative representation with Tikhonov regularization (KCRT) [27] classifier, relaxed sparse representation (RSR) classifier [28], [29], structure-

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aware CR with Tikhonov regularization (SaCRT) method [30], and feature-based collaborative representation [31].

To explore the intrinsic classification mechanism of SRC and CRC, Cai et al. [32] innovatively introduced the probability theory based on CRC. They analyzed the mechanism of CRC from the perspective of probability and then proposed a probabilistic collaborative representation classifier (ProCRC) method. ProCRC has achieved inspiring performance for face recognition and HSI analysis, which outperforms CRC [33]. A lot of improvements have been made on ProCRC to make it perform better in various applications [34].

The complex environment of wetlands and the limitation of spatial resolution of HSI make it difficult for a single classifier to classify wetland images very well. Some researchers have introduced ensemble learning methods and obtained the classification results with higher accuracy and stronger robustness [35], [36], [37]. Under these inspirations, ensemble learning becomes a new idea to deal with wetlands image classification. The main idea of ensemble learning is similar to the committee system, where multiple decision-makers make joint decisions to obtain optimal results [38]. Bagging and boosting are two main paradigms of ensemble learning according to the structure of classifiers.

Bagging paradigm is a kind of parallel structure with high efficiency in which classifiers are trained independently without affecting each other. After obtaining the basic classifiers' results, the final result is determined by the voting method [39]. Random forests (RFs) [40] and random subspaces (RSs) [41] are two representatives of bagging. Bagging can produce robust learning models with good noise immunity, especially for unstable algorithms. To encourage individual accuracy and diversity simultaneously within the ensemble, Rodriguez et al. [42] integrated principal component analysis (PCA) into each subspace and proposed a robust and stable rotating forest algorithm.

Boosting is a serial structure where the result of the previous classifier determines the setting of the next classifier [43], [44]. AdaBoost is the most typical Boosting algorithm, which can significantly improve the performance of weak learners [45]. In AdaBoost, the next classifier pays more focus on these samples that are difficult to classify in the previous classifier and intends to classify them correctly. Many classical boosting-based methods have also been proposed, among which gradient boosting decision tree (GBDT) [46] and its upgraded version, extreme gradient boosting (XGBoost) [47], [48] are widely used. Hosseiny et al. [49] combined ensemble learning with CNN to increase the transferability of the model and the reliability of the final results when dealing with the complex wetland image in Newfoundland.

The wetland environment is more complex than other landscapes. Three difficulties are common in HSIs classification: 1) the complexity of distribution of ground objects; 2) the similarity of spectrum of some objects; and 3) the difficulties and high costs in obtaining known samples [50]. These problems are more pronounced in wetland areas: 1) natural succession process of wetlands is disturbed due to human activities and the invasion of alien species. Many vegetations tend to alternate, leading to fragmentation of wetland ecosystems; 2) spatial resolution of HSIs is lower than that of MSIs and panchromatic images, which leads to a large number of mixed pixels in HSIs; 3) coastal wetlands have plenty of ground objects and the spectrum of each object are highly similar; and 4) wetland areas are often covered with primitive vegetation that has not been fully developed, making it difficult to obtain abundant known samples. These problems further increase the difficulty of wetland cover surveys.

In this article, the ProCRC method is first extended to its kernel function version. The proposed Kernel ProCRC method has excellent performance in classification tasks. Also, a novel method to define homogeneous neighborhoods with classifiers and morphological operations is developed. Then, a novel ensemble learning method that considers neighborhood and label information is proposed. Furthermore, to alleviate the difficulty brought by the small size samples, the idea of sample augmentation is added to ensemble learning, which further improves the performance of both basic classifiers and ensemble results.

The main contributions of this article are given as follows.

1) Considering high similarities in spectrum between heterogeneous objects of hyperspectral wetland datasets, the Kernel ProCRC method was proposed, which can map features with linear inseparability into a higher dimensional space and helps to separate similar samples without losing rich spectral information.
2) In order to deal with insufficient sample size, a novel pseudo-labeling strategy was designed. Different from traditional methods which were merely based on spectrum, the proposed method joints classifier and spatial information together. Morphological operation was imposed on neighboring samples to ensure the reliability of predicted pseudo labels.
3) To consider the diversity of dictionaries, two novel sampling methods, equal scale random and augmented scale composite sampling, were proposed after obtaining pseudo labels, which can capture equal samples from original and expanded dictionary without information loss and select disproportionate samples from two training sets. The proposed methods can take diversity and quality into consideration simultaneously. The size-flexible extraction of training samples enhanced the flexibility of classification.

The rest of this article is organized as follows. Section II describes the related works. In Section III, the detailed descriptions of Kernel ProCRC algorithm and two ensemble strategies are provided. In Section IV, the comparative performance of our methods with several classical and advanced HSI classification techniques and ensemble learning methods is presented. The effectiveness evaluation, parameters analysis, and ablation study are discussed in Section V. Finally, conclusions are drawn in Section VI.

II. Related Work

The representation based classifiers are based on the assumption that high-dimensional data mainly lie in low-dimensional subspace; thus, all samples (query samples) can be represented by linear combinations of representative

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samples (dictionary atoms). The samples with known labels are often used directly as the atoms, that is, the training samples mentioned in this article. Let training dataset $\mathbf{X} = [\mathbf{x}_1, \mathbf{x}_2, \ldots, \mathbf{x}_N] = [\mathbf{X}_1, \mathbf{X}_2, \ldots, \mathbf{X}_C] \in \mathbb{R}^{d\times N}$ represent the dictionary containing N atoms, where each atom is a d-dimensional training sample. Note that C refers to the total number of classes. Simultaneously, these atoms are selected from C different classes and $\mathbf{X}_c = [\mathbf{x}_1^c, \mathbf{x}_2^c, \ldots, \mathbf{x}_{N_c}^c] \in \mathbb{R}^{d\times N_c}$ denotes the training set of atoms belonging to the c th class, where $c = [1, 2, \ldots, C]$. Here, N_c is the number of atoms from the c th classes and $\sum_{c=1}^{C} N_c = N$, and $\mathbf{y} \in \mathbb{R}^d$ denotes the query sample awaiting classification. The set consisting of all query samples is denoted as $\mathbf{Y}$.

Hence, the mentioned assumption can be formulated as

$$\mathbf{y} = \mathbf{X}\boldsymbol{\alpha} \tag{1}$$

where $\boldsymbol{\alpha}$ represents the coefficient of the linear representation, which can also be interpreted as the proportions of training samples in the composition of $\mathbf{y}$. From another perspective, the representation based classification methods aim to find a common point $\mathbf{X}\boldsymbol{\alpha}$, which is similar to query sample $\mathbf{y}$ in the subspace spanned by the training samples $\mathbf{X}$.

A. SRC and CRC

For SRC, the main idea is to use few training samples to represent testing samples. Two important constraints are imposed: 1) the linear combination of training samples is similar to the testing sample, which is the reconstruction residual constraint, and 2) few training samples are selected in the linear combination, which is the sparse constraint [51]. It is implemented in two steps: calculating the representation coefficient $\boldsymbol{\alpha}$ and classifying. The representation coefficient can be calculated as

$$\boldsymbol{\alpha} = \arg\min_{\boldsymbol{\alpha}} \left(\|\mathbf{y} - \mathbf{X}\boldsymbol{\alpha}\|_2^2 + \lambda \|\boldsymbol{\alpha}\|_0 \right) \tag{2}$$

where the parameter λ is artificially set to maintain the balance between residual and sparsity.

Recent research showed that "collaboration" rather than "competition" among dictionary atoms is the key to the success of SRC on classification and sparsity is no longer emphasized when solving coefficients. Testing samples can be collectively represented by atoms of all classes. In the coefficient optimization formula, the l_0-norm is replaced by the l_2-norm, and consequently, a closed solution can be easily obtained by

$$\boldsymbol{\alpha} = \left(\mathbf{X}^T\mathbf{X} + \lambda\mathbf{I}\right)^{-1}\mathbf{X}^T\mathbf{y} \tag{3}$$

where $\mathbf{I}$ is an identity matrix. Compared with SRC, CRC is computationally efficient due to its closed-form solution. After representation coefficient $\boldsymbol{\alpha}$ is calculated, $\mathbf{y}$ can be directly composed of training samples by the following formula:

$$\mathbf{y} \approx \mathbf{X}\boldsymbol{\alpha} = \mathbf{X}_1\boldsymbol{\alpha}_1 + \cdots + \mathbf{X}_c\boldsymbol{\alpha}_c + \cdots + \mathbf{X}_C\boldsymbol{\alpha}_C \tag{4}$$

where $\boldsymbol{\alpha}_c$ denotes the partial coefficients in $\boldsymbol{\alpha}$ that correspond to dictionary atoms belonging to the c th class; in other words, $\boldsymbol{\alpha} = [\boldsymbol{\alpha}_1, \boldsymbol{\alpha}_2, \ldots, \boldsymbol{\alpha}_c, \ldots, \boldsymbol{\alpha}_C] \in \mathbb{R}^{N\times 1}$.

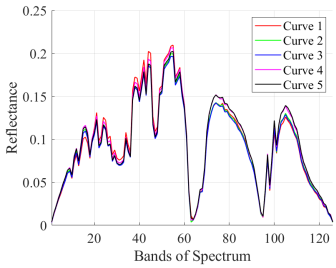


Fig. 1. Five similar real spectral curves.

Finally, a specific classification rule is required to assign a unique label to each query sample. Intuitively, the contribution made by atoms of each class in composing $\mathbf{y}$ is positively correlated with the probability that $\mathbf{y}$ belongs to each class. In consequence, the classification rule can be described as

$$\text{label}(\mathbf{y}) = \arg\min_{c=1,2,\ldots,C} \frac{\|\mathbf{y} - \mathbf{X}_\mathbf{c}\boldsymbol{\alpha}_\mathbf{c}\|_2^2}{\|\boldsymbol{\alpha}_\mathbf{c}\|_2^2} \tag{5}$$

where label($\mathbf{y}$) means the predicted label of query sample $\mathbf{y}$.

B. ProCRC

The ProCRC algorithm utilizes the Euclidean distance between samples and Gaussian function so that the probability that query sample $\mathbf{y}$ belongs to the c th class $P(\text{class}(\mathbf{y}) = c)$ can be clearly defined and calculated by

$$P(\text{label}(\mathbf{y}) = c) \propto \exp\big(-\big(\|\mathbf{y} - \mathbf{X}\boldsymbol{\alpha}\|_2^2 + \lambda\|\boldsymbol{\alpha}\|_2^2 + \gamma\|\mathbf{X}\boldsymbol{\alpha} - \mathbf{X}_c\boldsymbol{\alpha}_c\|_2^2\big)\big). \tag{6}$$

The main difference between ProCRC and CRC is the computation of the representation coefficient $\boldsymbol{\alpha}$. In ProCRC, the following optimization formula and its closed solution are used to calculate the representation coefficients:

$$\boldsymbol{\alpha} = \arg\min_{\boldsymbol{\alpha}} \begin{pmatrix} \|\mathbf{y} - \mathbf{X}\boldsymbol{\alpha}\|_2^2 + \lambda\|\boldsymbol{\alpha}\|_2^2 \\ +\gamma \sum_{c=1,2,\ldots,C} \|\mathbf{X}\alpha - \mathbf{X}_c\boldsymbol{\alpha}_c\|_2^2 \end{pmatrix} \tag{7}$$

$$\boldsymbol{\alpha} = \left(\mathbf{X}^T\mathbf{X} + \gamma \sum_{c=1,2,\ldots,C} \left(\overline{\mathbf{X}_c'}\right)^T \overline{\mathbf{X}_c'} + \lambda\mathbf{I}\right)^{-1} \mathbf{X}^T\mathbf{y} \tag{8}$$

where X_c' is a matrix with the same size as X, while only the samples of X_c will be assigned to X_c' at their corresponding locations. Also, $\bar{X}_c' = X - X_c'$. γ is a parameter set artificially, which is used to balance the proportions of the three items.

From the results, the addition of the third item adjusts the global $\boldsymbol{\alpha}$ through $\boldsymbol{\alpha}_c$ of each class. In this way, the combination of samples corresponding to each class $\mathbf{X}_c\boldsymbol{\alpha}_c$ is closer to that of global samples $\mathbf{X}\boldsymbol{\alpha}$. In a sense, the third item gives the representation of each class a clear direction. They are approaching the global point $\mathbf{X}\boldsymbol{\alpha}$. Thereby, the consistency of the representation is enhanced.

In order to show the difference between CRC and ProCRC intuitively, some comparative experiments are designed. Five samples with similar spectrums (shown in Fig. 1) are selected. These samples were put into two classifiers, and their representation coefficients were calculated by (3) and (7). The distributions of calculated coefficients are shown in Fig. 2. Notably, compared with CRC, the coefficients calculated by

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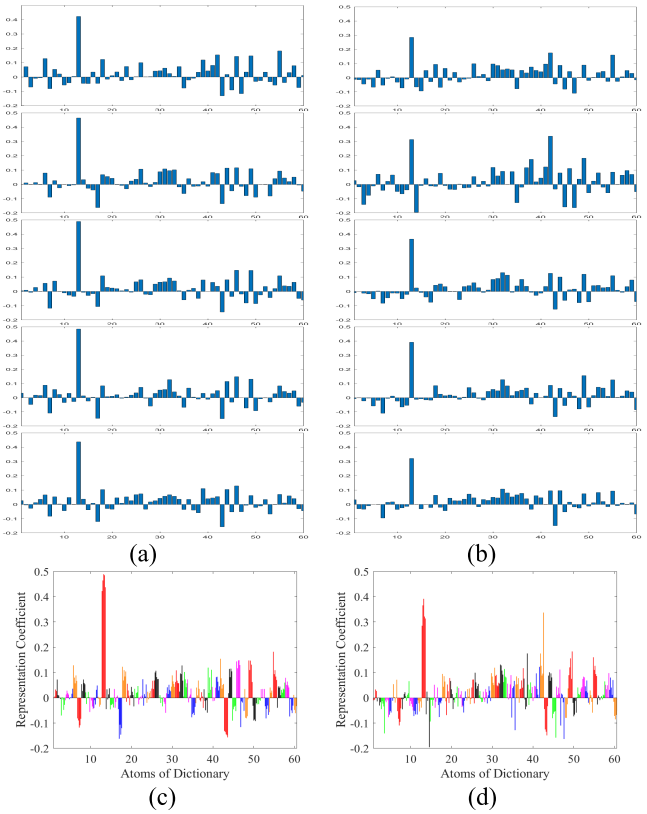


Fig. 2. Histogram of coefficients. (a) and (b) Five groups of coefficients calculated by ProCRC and CRC, respectively. (c) and (d) Coefficients in (a) and (b) are placed in the same histogram, respectively, where different colors indicate the coefficients corresponding to different atoms.

ProCRC are more consistent. Also, the differences between the two groups of coefficients are calculated by

$$\text{var} = \sum_{i=1}^{5} (\boldsymbol{\alpha}_i - \hat{\boldsymbol{\alpha}})^2 \tag{9}$$

where $\hat{\boldsymbol{\alpha}}$ represents the mean of $\boldsymbol{\alpha}_i$, $i = 1, \ldots, 5$. For CRC, the variance is 0.4132, whereas for ProCRC, the variance is 0.2001. The results show that the representation coefficients obtained by ProCRC are more consistent.

The reconstruction results of different classes in CRC and ProCRC were also compared. In the Purdue Campus dataset, a sample (its corresponding ground truth is tree) is selected and classified in CRC and ProCRC. The spectrum of test sample and reconstruction spectrums of each class $\mathbf{X}_c\boldsymbol{\alpha}_c$ are shown in Fig. 3. Because of the existence of the third item in ProCRC, the reconstruction spectrums of single classes are closer to the test sample. In other words, all classes are trying to combine dictionary atoms approaching test sample. It avoids the reverse distributions (grass and soil) in Fig. 3(a). It also helps to get the correct and reliable classification results.

The ProCRC is also different from CRC in the classification rules, which is expressed as

$$\text{label}(\mathbf{y}) = \arg \min_{c=1,2,\ldots,C} \frac{\|\mathbf{X}\boldsymbol{\alpha} - \mathbf{X}_c\boldsymbol{\alpha}_c\|_2^2}{\|\boldsymbol{\alpha}_c\|_2^2} \tag{10}$$

where $\mathbf{y}$ is replaced by the common point $\mathbf{X}\boldsymbol{\alpha}$ when measuring the Euclidean distances. This change makes the classification

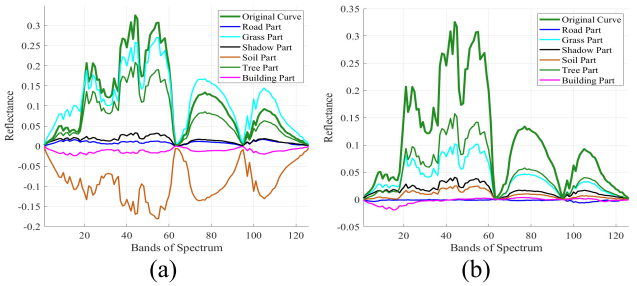


Fig. 3. Reconstruction result of different classes. Different colors represent different classes. (a) CRC. (b) ProCRC.

results unaffected by the diversity of the query samples caused by spectral heterogeneity.

C. Bagging

Two key requirements for ensemble learning are accuracy and diversity [36]. There are two main strategies to improve the diversity of classifiers: for heterogeneous ensemble, different types of classifiers can be selected as basic classifiers, while for homogeneous ensemble, different results are produced by different training data and features. Bagging, short for bootstrap aggregating, is an intuitive, simple and classic ensemble learning method. The main process of this method is to perform random sampling with replacement in the original training set to get N samples. The selected samples are used as a training set to learn a classifier. After T repetitions, T classifiers with different training sets can be obtained. Finally, the results of the T classifiers are voted by majority to make the final decision. The formula is

$$\text{label}(\mathbf{y}) = \text{mode}\left(\bigcup_{t=1}^{T} H_t(\mathbf{y})\right) \tag{11}$$

where $H(\mathbf{y})$ represents the predicted result of $\mathbf{y}$ and H_t represents that of the t th classifier. The mode $(\cdot)$ function can output the most frequent instances in the input data.

The training set corresponding to the t th classifier is defined as $\mathbf{X}_t$. Since the sampling method is with replacement, some samples in the original set $\mathbf{X}$ will appear multiple times in $\mathbf{X}_t$, while others do not appear. Suppose that there are n samples in $\mathbf{X}$. In each sampling, the probability that each sample is selected is $1/n$, so the probability of not selected is $(1-(1/n))$. For one of the samples, the probability that it is not selected after n times is $(1-(1/n))^n$. Also,

$$\lim_{n\to\infty}\left(1-\frac{1}{n}\right)^n = \frac{1}{e} \approx 0.368. \tag{12}$$

According to the theory above, about 36.8% of the samples in the original set will not be selected in a single subset and about 63.2% are selected. This method can reduce the impact of samples on the performance of the classifier and increase the robustness. If the classifier has high accuracy and is unstable, the bagging ensemble method tends to perform well.

III. Proposed Methods

A. Kernel ProCRC

Samples in HSI data are often linearly inseparable in the original spectral space, due to the complexity of the imaging

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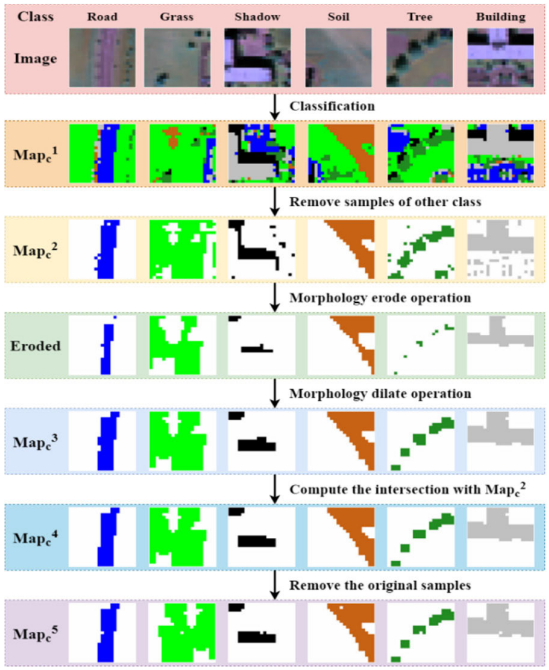


Fig. 4. Illumination of the process of obtaining neighborhood samples of the same class.

process of sensors and the existence of mixed pixels [52], [53]. They are often easy to separate in a higher dimensional space, but the calculation in that space is often difficult. The kernel function is the function of inputting vectors of the original space and outputting the dot product of vectors in high-dimensional space without explicitly identifying the nonlinear mapping [54], [55], [56]. The mapping is defined as Φ and the inner product operation of data in a high-dimensional space can be performed with a kernel function

$$K(\mathbf{d}_i, \mathbf{d}_j) = \Phi(\mathbf{d}_i)^T \Phi(\mathbf{d}_j). \tag{13}$$

However, some certain properties are necessary for kernel functions. Mercer's condition states that any semipositive definite function can be used as a kernel function. Gaussian kernel is chosen as the kernel function in this article because of its positive definiteness and full rank properties (which has been proved in [57] and [58]). The function can be written as

$$K(\mathbf{d}_i, \mathbf{d}_j) = e^{-\frac{\|\mathbf{d}_i - \mathbf{d}_j\|^2}{2\sigma^2}} \tag{14}$$

where $\|\cdot\|$ represents the modulus of the vector and σ is an artificially set parameter. According to the distribution of different data, the optimal value of σ is different.

In order to upgrade the ProCRC method to its kernel function version, it is necessary to calculate its representation coefficients $\boldsymbol{\alpha}$ in the feature space. The optimization formula and its closed solution are shown as follows:

$$\boldsymbol{\alpha} = \arg\min_{\boldsymbol{\alpha}} \left(\begin{array}{l} \|\Phi(\mathbf{y}) - \Phi(\mathbf{X})\boldsymbol{\alpha}\|_2^2 + \lambda\|\boldsymbol{\alpha}\|_2^2 \\ +\gamma \sum\limits_{c=1,2,\ldots,C} \|\Phi(\mathbf{X})\boldsymbol{\alpha} - \Phi(\mathbf{X}_c)\boldsymbol{\alpha}_c\|_2^2 \end{array} \right) \tag{15}$$

$$\boldsymbol{\alpha} = \left(K(\mathbf{X}, \mathbf{X}) + \gamma \sum_{c=1,2,\ldots,C} K\left(\mathbf{X}_c', \mathbf{X}_c'\right) + \lambda \mathbf{I} \right)^{-1} K(\mathbf{X}, \mathbf{y}). \tag{16}$$

Algorithm 1 Kernel ProCRC

Input: HSI data: $\mathbf{Y}$;
Training samples set: $\mathbf{X}$;
Label of training samples: $label(\mathbf{X})$;
Parameters: λ, γ, σ;

For each query sample y in Y

1. Calculate the inner product of the feature space data with (14) to get $K(X, X)$, $K(X, y)$ and $K(X_c', X_c')$;
2. Calculate the representation coefficient $\boldsymbol{\alpha}$ with (16);
3. Classify the query sample y with rule (17) and get its predicted label $label(\mathbf{y})$;

End

Output: Classification result of HSI: $label(\mathbf{Y})$;

After calculating $\boldsymbol{\alpha}$, use the following classification rules to classify the query samples:

$$\begin{aligned} &\text{label}(\mathbf{y}) \\ &= \arg\min_{c=1,2,\ldots,C} \frac{\|\Phi(\mathbf{X})\boldsymbol{\alpha} - \Phi(\mathbf{X}_c)\boldsymbol{\alpha}_c\|_2^2}{\|\boldsymbol{\alpha}_c\|_2^2} \\ &= \arg\min_{c=1,\ldots,C} \frac{\begin{array}{l}\boldsymbol{\alpha}^T\Phi(\mathbf{X})^T\Phi(\mathbf{X})\boldsymbol{\alpha} - \boldsymbol{\alpha}^T\Phi(\mathbf{X})^T\Phi(\mathbf{X}_c)\boldsymbol{\alpha}_c \\ -\boldsymbol{\alpha}_c^T\Phi(\mathbf{X})_c^T\Phi(\mathbf{X})\boldsymbol{\alpha} + \boldsymbol{\alpha}_c^T\Phi(\mathbf{X})_c^T\Phi(\mathbf{X}_c)\boldsymbol{\alpha}_c\end{array}}{\|\boldsymbol{\alpha}_c\|_2^2} \\ &= \arg\min_{c=1,\ldots,C} \frac{\begin{array}{l}\boldsymbol{\alpha}^T K(\mathbf{X}, \mathbf{X})\alpha + \boldsymbol{\alpha}_c^T K(\mathbf{X}_c, \mathbf{X}_c)\boldsymbol{\alpha}_c \\ -\boldsymbol{\alpha}^T K(\mathbf{X}, \mathbf{X}_c)\boldsymbol{\alpha}_c - \boldsymbol{\alpha}_c^T K(\mathbf{X}_c, \mathbf{X})\boldsymbol{\alpha}\end{array}}{\|\boldsymbol{\alpha}_c\|_2^2}. \end{aligned} \tag{17}$$

The pseudocode for the Kernel ProCRC method is detailed in Algorithm 1.

B. Local Neighborhood-Based Ensemble

The performance of basic classifiers is important to ensemble learning. However, bagging methods may not work well when representation based classifiers are used as basic classifiers because of the sampling methods, especially in the case of small size samples. In order to make full use of the existing samples, a new ensemble learning method based on spatial neighborhood and label information was proposed. Different from pattern recognition, the objects classified in remote sensing image represent the real objects on the surface. According to Tobler's First Law of Geography, near things are more related to each other [59]. Thus, spatial information plays an important role in remote sensing classification [60], [61]. It can eliminate discrete outliers in classification maps and improve the accuracy by leaps and bounds.

Before classifying, it is necessary to obtain samples of the same class as the known sample from its neighborhood and Fig. 4 shows the procedure step by step. First, define square neighborhoods Neighbor($\mathbf{X}$) with a side length R centered on the dictionary atoms $\mathbf{X}$. The images in the first line of Fig. 4 are false-color images of the neighborhoods. Map_c^1 is the classification map of neighborhoods

$$\text{Map}_c^1 = H(\mathbf{y} | \mathbf{y} \in \text{Neighbor}(\mathbf{X}_c)). \tag{18}$$

According to the classification results, samples that are the same as the central sample are extracted easily. The

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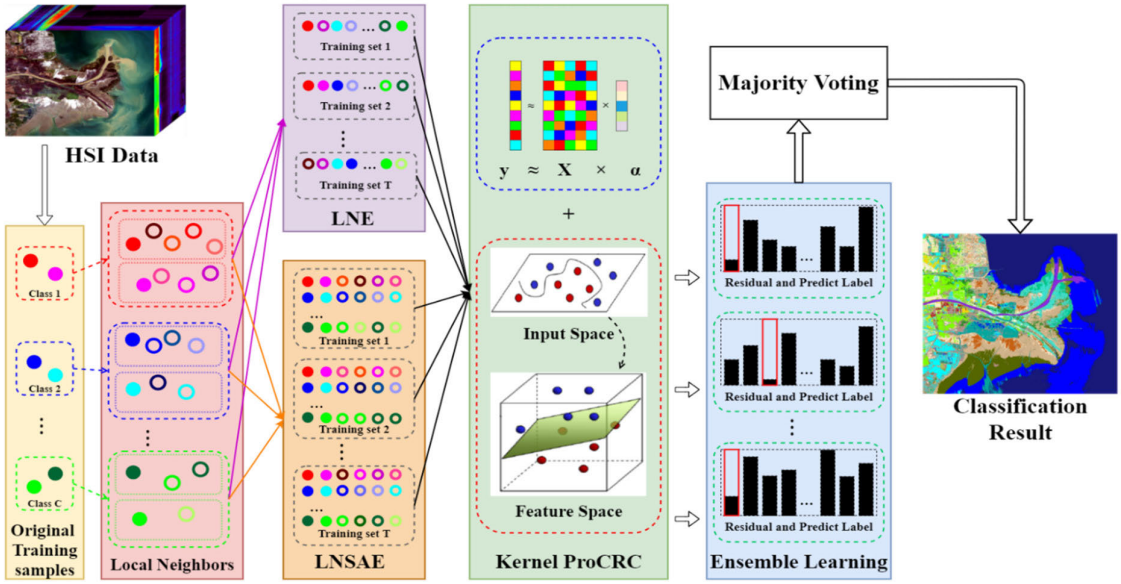


Fig. 5. Flowchart of the proposed KProCRC-LNSAE, including local neighbors, Kernel ProCRC methods, and two ensemble learning methods. Dots of different color series represent samples from different classes. Also, solid and hollow points represent the original samples and neighborhood samples, respectively.

classification map Map_c^2, which only contains samples of the c th class, is obtained form Map_c^1

$$\mathrm{Map}_c^2 = \left\{ \mathbf{y} | \mathbf{y} \in \mathrm{Map}_c^1 \,\&\, H(\mathbf{y}) = c \right\}. \tag{19}$$

Since the classification results are not completely accurate, compared with the small structures and isolated pixels, the main structures have a greater probability of representing the same object as the center samples $\mathbf{X}_c$. Thus, the morphological opening operations (morphological erosion is performed first, followed by morphological dilation) are used to remove the pixels that may not belong to the c th class. The formulas are shown as follows:

$$\mathbf{A} = \gamma_e(\mathbf{B}) = \left\{ x, y | e_{x,y} \subseteq \mathbf{B} \right\} \tag{20}$$

$$\mathbf{A} = \phi_e(\mathbf{B}) = \left\{ x, y | e_{x,y} \cap \mathbf{B} \neq \emptyset \right\} \tag{21}$$

$$\mathrm{Map}_c^3 = \phi_e\left(\gamma_e\left(\mathrm{Map}_c^2\right)\right) \tag{22}$$

where $\mathbf{A}$ and $\mathbf{B}$ represent the processed map and map before processing, respectively, e is a structural element, γ_e and ϕ_e represent morphological erosion and dilation operations with structure element e, respectively, and $e_{x,y}$ represents the structure element with center point at (x, y). Equation (20) is the morphological erosion that removes isolated points and very small structures, while (21) is the morphological dilation, which can restore the structure of the object itself and avoid excessive erosion. By (22), the map after morphological transformation Map_c^3 is obtained.

However, it may be overdilated and lead to some samples that are not c th class in Map_c^2 remaining in Map_c^3. Therefore, it is necessary to further calculate the intersection of Map_c^2 and Map_c^3 to ensure that the final retained neighborhood samples are of the same class as the central sample in the classification process. Thus, Map_c^4 can be obtained by

$$\mathrm{Map}_c^4 = \mathrm{Map}_c^2 \cap \mathrm{Map}_c^3. \tag{23}$$

The accuracy of the neighborhood samples determines the reliability of the ensemble learning method. There are three guarantees for it: 1) neighborhood samples have a high probability of belonging to the same class; 2) the method used for neighborhood classification is quite accurate; and 3) samples that might be misclassified will be removed by morphological operations. Therefore, there is a high probability that the neighborhood samples selected by our method are in the same class as the central samples.

After getting neighborhood samples of each training sample $\mathbf{x}$, the candidate samples of the c th class X_c^S are easily obtained

$$\mathbf{X}_c^S = \left\{ \mathbf{x} | \mathbf{x} \in \mathrm{Map}_c^4 \right\}. \tag{24}$$

Also, dictionaries with diversity can be constructed

$$\mathbf{X}_t = \bigcup_{c=1}^{C} \mathrm{sampling}_{N_c}^{W}\left(\mathbf{X}_c^S\right) \tag{25}$$

where $\mathrm{sampling}_{N_c}^{W}(\cdot)$ represents sampling N_c samples without replacement.

For the t th basic classifier, the corresponding dictionary $\mathbf{X}_t$ is put into the Kernel ProCRC method for classification. Finally, the results of each basic classifier are ensembled by majority voting to get the final predicted result.

Algorithm 2 shows the pseudocode (using MATLAB notations) for the local neighborhood-based ensemble (LNE) method step by step.

In our methods, spatial information is utilized in a novel way. Spatial information acts not on basic classifiers but on sample selection. Specifically, in the sample selection stage, spatial information is used to select candidate samples and construct diverse dictionaries. In the classification stage, Kernel ProCRC, the pixel-by-pixel classification method, is used without considering spatial information. In this way, the

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Algorithm 2 LNE

Input: HSI data: $\mathbf{Y}$;
Training samples set: $\mathbf{X}$;
Label of training samples: $label(\mathbf{X})$;
Basic classifier and its parameters (take Kernel ProCRC as an example here);
Number of classifiers: T;
Radius of the neighborhood R;
Morphological structural element: s;

Step 1: Find candidate samples

For $c = 1, 2, \ldots, C$

1. Define $Neighbor(\mathbf{x})$ for samples in $\mathbf{X}$.
2. Use Kernel ProCRC as classification method H to get Map_c^1 by (18).
3. Obtain Map_c^2 with only c th class samples by (19).
4. Acquire Map_c^3 with structural element s by (22).
5. Get Map_c^4 by (23).
6. Get X_c^S for each class by (24).

End

Step 2: Ensemble learning

For $t = 1, 2, \ldots, T$

1. Get the t th dictionary $\mathbf{X}_t$ by (25).
2. Put X_t into Kernel ProCRC for classification and obtain the t th classification result of Y $label_t(\mathbf{Y})$ by **Algorithm 1**.

End

Output: Getting the final predicted labels $label(\mathbf{Y})$ utilizing majority voting by (11).

proposed method utilizes the spatial information while preserving the detailed texture in the image simultaneously.

C. Local Neighborhood and Samples Augmentation-Based Ensemble

The previously proposed LNE method solves the problems of missing and duplicating of training samples, and however, some limitations still exist. The abundant samples produced by LNE are sometimes underutilized, so a more advanced ensemble learning method is proposed to introduce the idea of sample augmentation to the mentioned LNE method. Some larger dictionaries based on neighborhood samples are built. The number of atoms in dictionaries is a multiple of N, consisting of the original samples and randomly sampling ones from neighborhoods. The original samples ensure the accuracy, while the random sampling ones provide the diversity among dictionaries.

For purpose of avoiding the selection of original samples, they need to be selected in advance. Removing the central samples from Map_c^4, the neighborhood Map_c^5 is acquired

$$\mathrm{Map}_c^5 = \left\{ \mathbf{y} | \mathbf{y} \in \mathrm{Map}_c^4 \& \mathbf{y} \notin \mathbf{X}_c \right\}. \tag{26}$$

Also, the candidate samples for sampling of the c th class X_c^S are obtained in the same way

$$\mathbf{X}_c^S = \left\{ \mathbf{x} | \mathbf{x} \in \mathrm{Map}_c^5 \right\}. \tag{27}$$

Algorithm 3 LNSAE

Input: $\mathbf{Y}, \mathbf{X}, label(\mathbf{X}), T, R, s$ and Basic classifier and its parameters (the same as **Algorithm 2**)
Multiple of the number of training samples: M

Step 1: Find candidate samples

For $c = 1, 2, \ldots, C$

(Processes 1 to 5 like **Step1** in **Algorithm 2**).

6. Obtain Map_c^5 from Map_c^4 by (26).
7. Get $\mathbf{X}_c^S$ for each class by (27).

End

Step 2: Samples Augmentation Ensemble learning

For $t = 1, 2, \ldots, T$

1. Get the t th dictionary $\mathbf{X}_t$ by (28) and (29).
2. Put $\mathbf{X}_t$ into Kernel ProCRC for classification and obtain the t th classification result of $\mathbf{Y}$ $label_t(\mathbf{Y})$ by **Algorithm 1**.

End

Output: Getting the final predicted labels $label(\mathbf{Y})$ utilizing majority voting by (11).

For different classes, the number of candidate samples n_c varies. The sampling methods used are also different. When $n_c \geq (M-1) \times N_c$, it means that $\mathbf{X}_c^S$ is sufficient. Sampling without replacement is adopted to avoid duplicate samples. Otherwise, sampling with replacement is adopted to ensure the number of dictionary atoms. The random sampling of the t th dictionary $\mathbf{X}_t^{\mathrm{Ran}}$ can be described as

$$\mathbf{X}_t^{\mathrm{Ran}} = \bigcup_{c=1}^{C} \begin{cases} \mathrm{sampling}_{N_c}^{W}\left(\mathbf{X}_c^S\right), & n_c \geq (M-1) \times N_c \\ \mathrm{sampling}_{N_c}^{R}\left(\mathbf{X}_c^S\right), & n_c < (M-1) \times N_c \end{cases} \tag{28}$$

where $\mathrm{sampling}_{N_c}^{R}(\cdot)$ represents sampling N_c samples with replacement. Adding original samples, the dictionary of the t th classifier $\mathbf{X}_t$ is obtained

$$\mathbf{X}_t = \mathbf{X} \cup \mathbf{X}_t^{\mathrm{Ran}} \tag{29}$$

with the expanded dictionaries, and the classification and ensemble learning process is the same as Algorithm 2. The details of the proposed local neighborhood and samples augmentation-based ensemble (LNSAE) method are shown gradually by the pseudocode in Algorithm 3.

The flowchart of the proposed methods is shown in Fig. 5.

IV. Experiments and Results

In this section, three real-world datasets are first introduced, and the performance of the proposed methods is evaluated through a series of experiments.

A. Experimental Hyperspectral Data

The datasets used in this article are the imaging data of three typical wetland areas by the GF-5 satellite. The data collected by GF-5 contain 330 bands with a spatial resolution of 30 m. There are 150 visible and near-infrared (VNIR, 0.39–1.03 μm) bands with a spectral resolution of 5 nm and 180 shot-wave infrared (SWIR, 1.0–2.5 μm) bands with a spectral resolution of 10 nm. The original data are scaled in the range of [0, 1]

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TABLE I
TWENTY-ONE CLASSES OF THE YELLOW RIVER DELTA DATASET

Class	Ground object	Number of samples
1	Aquaculture	393
2	Seep Sea	796
3	Soybean	110
4	Rice	190
5	Building	83
6	Maize	96
7	Broomcorn	95
8	Locust	211
9	Spartina	200
10	Shallow Sea	936
11	Mud Flat	553
12	River	469
13	Suaeda salsa	361
14	Reed	240
15	Salt marsh	595
16	Intertidal saltwater	454
17	Tamarix	133
18	Pond	377
19	Flood plain	68
20	Freshwater herbaceous marsh	72
21	Aquatic vegetation	39
Total		6471

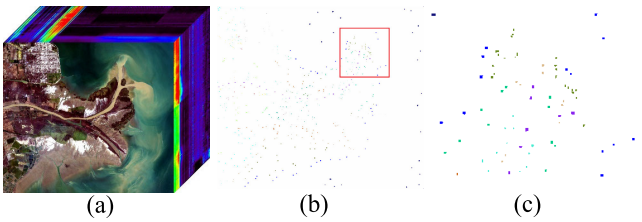


Fig. 6. Maps of Yellow River Delta dataset. (a) False-color image. (b) Ground truth. (c) Zoom of part of (b) to highlight the distribution.

for the convenience and efficiency of calculation. These hyperspectral remote sensing datasets are public datasets and can be downloaded from the China Landsat Data Center and Grid Platform of High Resolution Earth Observation System.

1) Yellow River Delta: It was acquired over estuary of Yellow River in Dongying, Shandong province, China. The image consists of 1185 × 1324 pixels and covers 285 spectral bands (45 noisy bands have been removed). As shown in Table I, 21 primary classes of ground objects are contained in this area, and the false-color composite image and ground reference image are shown in Fig. 6(a) and (b), respectively. In order to display the distribution of ground truth more clearly, the zoom map of the part in red box in Fig. 6(b) is shown in Fig. 6(c).

2) Yangtze River Delta: It is an image of the Nantong part of the Yangtze River Delta, eastern of Jiangsu province, China. The image size is 1160 × 1290; 268 bands are taken after removing 62 noisy bands. The image data contain nine types of typical ground objects, and the detailed sample information is shown in Table II. The images are shown in Fig. 7.

3) Dafeng Natural Reserve: It is the HSI of Dafeng Nature Reserve, located on the southern shore of the Yellow Sea in Yancheng, Jiangsu province, China, with the size of 986 × 632. Seventy-four noisy bands were removed, and 256 high-quality bands were retained. There are mainly nine types of ground objects contained in this area and details

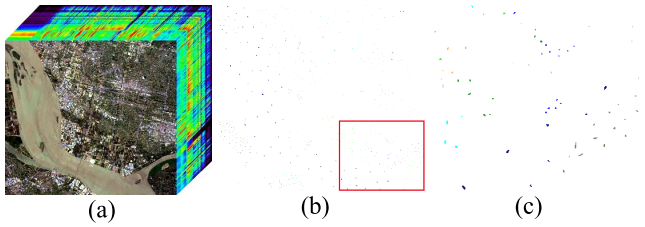


Fig. 7. Maps of Yangtze River Delta dataset. (a) False-color image. (b) Ground truth. (c) Zoom of part of (b) to highlight the distribution.

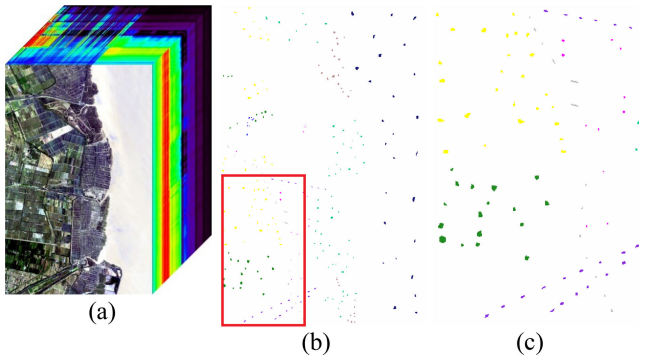


Fig. 8. Maps of Dafeng Natural Reserve dataset. (a) False-color image. (b) Ground truth. (c) Zoom of part of (b) to highlight the distribution.

TABLE II
NINE CLASSES OF THE YANGTZE RIVER DELTA DATASET

Class	Ground object	Number of samples
1	River	1273
2	Trees	575
3	Bare Land	200
4	Farmland	189
5	Lake	137
6	Mud Flat	476
7	Flood Plain	435
8	Impervious Surface	1046
9	Plant	455
Total		4786

TABLE III
NINE CLASSES OF THE DAFENG NATURAL RESERVE DATASET

Class	Ground object	Number of samples
1	Sea	1020
2	Mud Flat	528
3	Road	150
4	Buildings	89
5	Farmland	918
6	River	276
7	Trees	724
8	Aquiculture	746
9	Lake	79
Total		4530

are shown in Table III. The images of these data are shown in Fig. 8.

B. Experimental Setup

To objectively evaluate the proposed methods, popular classification methods are selected for comparison. Extreme learning machine (ELM) [62], 3-D CNN [16], ProCRC [32], and kernel elastic net LRR (KENLRR) [19] are chosen as the contrast single classifiers. Rotation forest (RoF) [42], XGBoost [48], and TCRC-Bagging [43] algorithm are chosen for ensemble learning. To simulate the small size sample situation, ten known samples are randomly selected from

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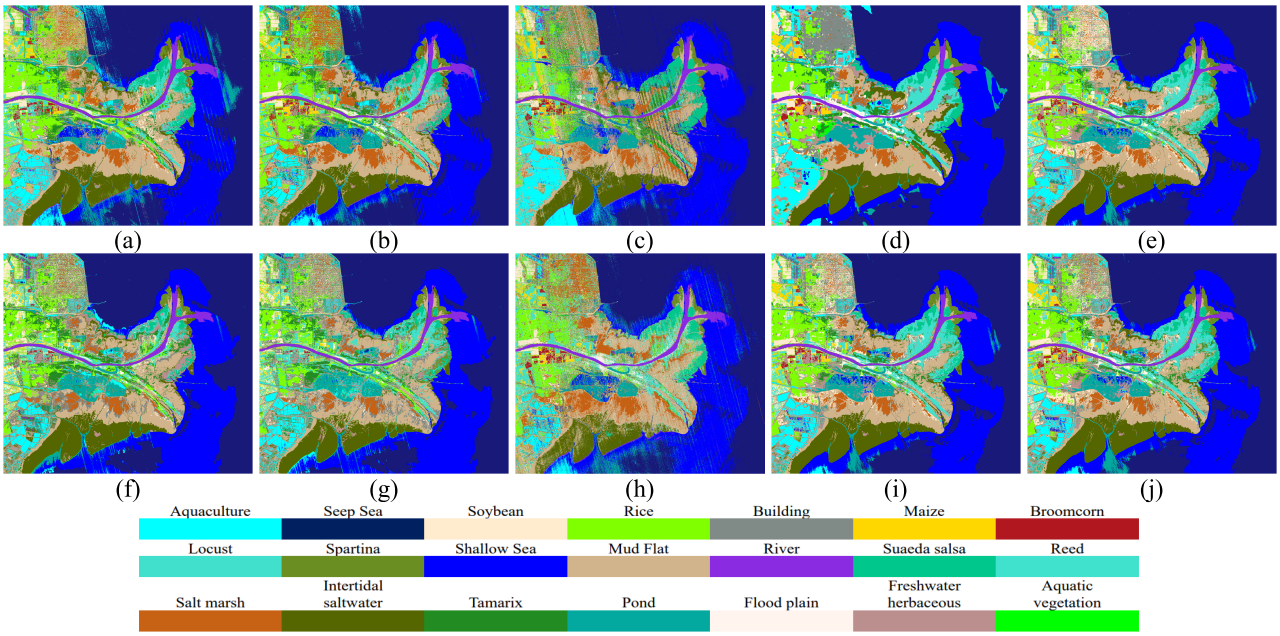


Fig. 9. Classification maps of Yellow River Delta. (a) ELM. (b) 3-D CNN. (c) ProCRC. (d) KENLRR. (e) Kernel ProCRC. (f) RoF. (g) XGBoost. (h) TCRC-Bagging. (i) KProCRC-LNE. (j) KProCRC-LNSAE. (k) Class legend.

TABLE IV
PARAMETERS SETTING

Methods \ Dataset		Yellow River Delta	Yangtze River Delta	Dafeng Natural Reserve
Kernel ProCRC	λ	1e-3	1e-4	1e-3
	γ	1e-4	1e-5	1e-5
	σ	0.1	0.1	0.2
KProCRC-LNE	R	7	13	13
KProCRC-LNSAE	R	7	13	11
	M	3	3	3

each class as the training set. The optimal parameters of the proposed methods are listed in Table IV. The parameters of the rest of the methods are adjusted to the optimum as far as possible. For ensemble learning, the number of basic classifiers T is set to 5.

C. Classification Performance

To reflect the performance of these methods, the overall accuracy (OA), average accuracy (AA), kappa coefficient (Kappa), and individual class accuracy (CA) are calculated. Each group of experiments is conducted ten times with randomly selected samples, and the average results were taken as the final results to offset the accidental error. The values in parentheses after accuracies are the corresponding variances. In order to compare the efficiency of various classification methods, the running time was considered. All of them are implemented using MATLAB environment on a 2.60-GHz CPU.

The performances of the proposed methods and contrast methods are listed in Tables V–VII and the best results are emphasized in bold. The corresponding classification maps are shown in Figs. 9–11.

For Yellow River Delta data, the accuracy of the KProCRC-LNSAE is the highest, reaching 97.29%, which is 7.28% higher than that of the ProCRC. The accuracy of Kernel ProCRC is 95.81%, which increased by 5.80% compared with ProCRC. The OA of KProCRC-LNE is second to the highest (96.32%). The two methods mentioned above can accurately identify most of the ground objects with high CA, such as broomcorn, locust, and mud flat. From the classification map in Fig. 9, some basic classifiers, such as ELM, 3-D CNN, and ProCRC, have streaks and noise in the map, while for the ensemble learning method, these phenomena are greatly improved. KENLRR can get high accuracy (95.66%). However, there are abundant homogeneous patches due to the use of spatial information. It will force the neighborhood to be constrained to the same class and come at the cost of losing detailed texture information. The proposed methods preserve these details well.

For Yangtze River Delta data, the accuracy of KProCRC is 89.78%, while that of KProCRC-LNSAE (highest among all methods) is 91.84%, improved by 5.56% than ProCRC. Comparing ProCRC and KProCRC, the use of kernel function can greatly enhance the classification ability. It can be found from the classification maps that with ProCRC, RoF, XGBoost, and TCRC-Bagging methods, trees and impervious surface on land can be misclassified to bare land. The proposed methods do not suffer from it, resulting in clear maps. Almost all methods misclassified some water in the Yangtze River as flood plain and bare land. This may be due to the fact that there are more sediment and soil components in the water flow, resulting in its spectrum similar to flood plain and bare land.

The highest OA of Dafeng Natural Reserve is 95.46%, which is realized by KProCRC-LNSAE. The OA of KProCRC is 93.68%, improved by 3.56% compared with ProCRC. KENLRR has the accuracy of 95.28% and is only less than KProCRC-LNSAE, while the corresponding map lost lots of detail. In contrast, the three proposed methods can not only

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TABLE V
CLASSIFICATION ACCURACY (%) FOR THE YELLOW RIVER DELTA DATASET

Class	Train	Validation	ELM	3D CNN	ProCRC	KENLRR	Kernel ProCRC	RoF	XGBoost	TCRC-Bagging	KProCRC-LNE	KProCRC-LNSAE
1	10	383	77.15(12.21)	79.47(11.58)	90.67(9.52)	**95.88(6.35)**	88.27(10.55)	91.46(14.19)	92.55(19.22)	94.80(6.74)	95.81(5.24)	95.45(3.22)
2	10	786	98.43(0.25)	97.34(3.34)	97.19(1.27)	99.69(0.95)	99.52(1.02)	95.30(5.48)	97.79(1.52)	99.68(1.24)	99.55(0.67)	**99.87(0.51)**
3	10	100	68.50(7.01)	67.37(14.24)	69.53(12.37)	**85.74(10.78)**	75.03(9.66)	67.34(14.46)	70.51(20.75)	67.70(16.31)	80.87(5.21)	83.58(8.69)
4	10	180	94.18(3.22)	95.16(6.26)	86.88(8.56)	93.97(6.34)	93.93(4.85)	86.15(9.45)	85.16(15.34)	86.28(6.91)	94.98(4.12)	**96.79(3.22)**
5	10	73	96.85(6.02)	99.27(0.93)	92.09(3.56)	78.03(6.83)	99.84(0.51)	70.20(9.53)	83.55(7.53)	90.03(7.16)	98.98(0.48)	**99.66(0.27)**
6	10	86	91.44(6.21)	95.27(6.66)	88.19(7.65)	99.38(1.02)	98.75(0.94)	92.30(7.45)	94.90(3.12)	89.58(8.71)	**99.76(0.21)**	99.74(0.01)
7	10	85	92.36(9.15)	96.14(13.41)	94.84(2.54)	96.98(1.77)	98.16(6.53)	80.07(8.85)	85.70(3.05)	96.40(2.17)	**100.00(0.00)**	**100.00(0.00)**
8	10	201	99.17(0.21)	97.88(5.25)	96.76(3.42)	**100.00(0.00)**	99.95(0.06)	99.12(0.45)	99.70(0.43)	97.47(3.74)	**100.00(0.00)**	**100.00(0.00)**
9	10	190	98.26(1.26)	**99.73(0.25)**	94.47(11.43)	95.60(7.52)	96.75(3.24)	97.13(1.17)	98.94(0.72)	97.28(3.25)	95.93(4.13)	96.74(1.27)
10	10	926	90.00(4.25)	95.01(2.75)	91.86(9.73)	96.85(8.62)	94.70(11.42)	97.28(3.91)	97.19(6.49)	94.85(8.42)	96.43(3.91)	**98.29(5.02)**
11	10	543	98.53(0.55)	98.60(2.94)	92.21(6.41)	99.10(1.24)	99.28(2.54)	96.23(4.62)	97.97(6.48)	92.79(7.84)	99.52(0.84)	**99.66(0.49)**
12	10	459	98.29(1.96)	98.68(2.99)	95.44(5.72)	**99.60(0.96)**	98.55(2.71)	96.58(5.82)	97.75(4.63)	95.11(6.65)	99.35(0.11)	99.52(0.09)
13	10	351	99.27(0.21)	99.39(0.84)	90.70(7.51)	**100.00(0.00)**	99.58(0.26)	98.92(4.33)	99.32(0.63)	92.86(8.42)	**100.00(0.00)**	99.65(0.98)
14	10	230	**94.99(0.33)**	93.74(5.63)	85.88(11.75)	87.07(21.53)	93.46(19.63)	88.64(15.52)	91.53(8.34)	90.41(0.19)	93.21(0.02)	93.56(0.03)
15	10	585	99.60(0.59)	99.77(0.99)	96.77(6.23)	**100.00(0.00)**	99.98(0.09)	99.92(0.34)	99.90(0.45)	94.46(7.34)	99.86(0.93)	99.97(0.02)
16	10	444	98.72(1.53)	**99.36(1.01)**	97.58(7.93)	98.07(4.51)	99.80(0.26)	97.51(6.41)	97.81(3.72)	98.40(0.72)	99.86(0.33)	99.86(0.16)
17	10	123	85.05(4.27)	92.43(5.34)	68.22(14.85)	89.05(19.60)	91.58(7.32)	79.98(22.96)	86.43(14.92)	82.61(21.60)	92.49(4.69)	**96.05(2.19)**
18	10	367	72.20(19.26)	73.04(21.22)	75.06(14.30)	86.47(20.10)	88.54(12.77)	75.36(32.15)	76.78(29.51)	79.83(19.67)	81.72(5.44)	**88.47(4.84)**
19	10	58	70.11(4.18)	54.23(20.24)	54.54(23.90)	**79.94(14.25)**	73.27(23.67)	69.00(24.35)	61.25(9.47)	43.72(17.92)	73.74(32.98)	62.83(15.30)
20	10	62	83.47(19.64)	91.78(15.32)	57.87(24.56)	86.33(10.20)	94.25(7.07)	84.00(13.92)	85.86(12.68)	81.58(18.36)	**95.44(8.70)**	95.17(1.35)
21	10	29	91.71(17.26)	96.04(15.79)	69.61(10.95)	95.75(11.93)	96.93(5.21)	94.15(7.12)	92.26(3.87)	85.61(21.27)	82.58(9.19)	**98.00(0.97)**
OA (%)			92.42(7.95)	93.28(6.15)	90.01(11.63)	95.66(2.41)	95.81(1.39)	92.33(6.51)	93.75(9.54)	91.32(14.37)	96.32(1.45)	**97.29(0.59)**
AA (%)			90.39(6.47)	91.41(6.36)	85.07(22.59)	93.50(6.83)	94.29(2.15)	88.41(12.32)	90.13(10.26)	88.16(19.32)	94.29(2.15)	**95.37(1.35)**
Kappa (%)			91.78(11.31)	92.69(9.94)	89.16(13.52)	95.30(3.68)	95.45(1.74)	91.68(14.26)	93.21(8.34)	90.60(13.19)	96.00(1.98)	**97.05(0.66)**
Time (s)			120.77	199.28	144.00	152.98	188.13	253.51	267.52	379.90	407.36	423.61

TABLE VI
CLASSIFICATION ACCURACY (%) FOR THE YANGTZE RIVER DELTA DATASET

Class	Train	Validation	ELM	3D CNN	ProCRC	KENLRR	Kernel ProCRC	RoF	XGBoost	TCRC-Bagging	KProCRC-LNE	KProCRC-LNSAE
1	10	1263	99.60(0.40)	99.80(0.91)	97.53(1.98)	97.39(3.73)	99.75(0.09)	99.55(1.03)	99.43(0.97)	96.44(2.19)	**99.85(0.43)**	99.82(0.56)
2	10	565	72.72(15.87)	82.67(6.87)	77.89(14.46)	84.17(7.84)	86.70(17.91)	76.42(18.77)	77.96(12.46)	78.90(14.11)	86.56(11.75)	**90.30(0.66)**
3	10	190	77.30(14.81)	81.07(14.63)	73.23(23.86)	89.30(15.51)	87.91(8.01)	47.03(32.57)	52.40(10.48)	83.92(4.96)	86.93(4.32)	**90.86(3.41)**
4	10	179	77.18(15.41)	91.33(6.17)	88.01(8.40)	90.99(5.59)	88.28(6.99)	74.74(15.82)	77.57(11.43)	**94.78(2.33)**	90.66(5.87)	90.76(2.24)
5	10	127	49.57(26.15)	53.05(21.80)	63.62(7.16)	64.86(10.46)	60.01(17.23)	53.53(25.96)	51.49(14.51)	**72.59(9.86)**	60.39(11.13)	67.62(17.05)
6	10	466	59.27(19.32)	68.85(18.85)	62.64(14.05)	71.02(10.04)	69.31(18.63)	62.05(20.56)	63.00(18.03)	63.61(21.94)	69.01(12.37)	**72.03(8.63)**
7	10	425	79.12(15.90)	91.87(9.88)	91.15(7.49)	85.46(10.57)	91.09(8.49)	83.99(12.61)	89.35(14.27)	**92.86(4.14)**	91.18(5.04)	91.51(5.02)
8	10	1036	83.94(6.59)	93.37(6.26)	90.09(14.61)	84.02(15.57)	94.48(4.31)	91.97(8.43)	88.53(8.52)	83.83(6.94)	95.97(1.27)	**96.71(0.83)**
9	10	445	**100.00(0.00)**	99.92(0.02)	99.73(0.34)	99.54(0.13)	99.98(0.04)	99.18(0.09)	99.39(0.03)	99.88(0.05)	99.98(0.02)	99.95(0.01)
OA (%)			81.50(7.61)	88.82(8.57)	86.28(11.49)	86.13(3.84)	89.78(6.63)	81.99(9.68)	82.62(13.57)	85.86(11.77)	90.15(1.37)	**91.84(0.59)**
AA (%)			77.63(7.53)	84.66(7.07)	82.65(12.75)	85.20(4.11)	86.39(6.58)	76.50(7.90)	77.68(12.34)	85.20(15.20)	86.73(1.65)	**88.84(1.67)**
Kappa (%)			77.97(9.20)	86.66(5.34)	83.61(11.27)	83.45(4.00)	87.83(5.08)	78.76(11.82)	79.43(10.88)	83.05(17.51)	88.28(1.47)	**90.26(0.36)**
Time (s)			135.06	211.38	157.41	163.73	200.49	309.05	317.53	423.64	469.75	493.01

TABLE VII
CLASSIFICATION ACCURACY (%) FOR THE DAFENG NATURE RESERVE DATASET

Class	Train	Validation	ELM	3D CNN	ProCRC	KENLRR	Kernel ProCRC	RoF	XGBoost	TCRC-Bagging	KProCRC-LNE	KProCRC-LNSAE
1	10	1010	99.97(0.02)	100.00(0.00)	99.91(0.00)	**100.00(0.00)**	99.99(0.01)	99.94(0.02)	**100.00(0.00)**	99.91(0.05)	**100.00(0.00)**	**100.00(0.00)**
2	10	518	89.66(7.30)	94.11(9.18)	91.25(10.15)	97.94(4.49)	93.22(6.69)	85.30(15.33)	78.88(16.43)	90.26(5.72)	96.24(1.36)	**98.97(0.56)**
3	10	140	41.79(17.03)	71.22(10.87)	54.42(17.60)	56.15(18.81)	61.90(15.55)	38.99(14.32)	30.51(13.24)	43.69(17.93)	65.03(15.48)	**72.09(11.71)**
4	10	79	99.85(0.10)	100.00(0.00)	94.12(3.68)	96.00(2.16)	99.75(0.22)	97.71(0.48)	**100.00(0.00)**	**100.00(0.00)**	99.75(0.13)	97.95(0.28)
5	10	908	95.73(7.60)	96.09(7.39)	93.57(6.36)	97.20(3.26)	96.37(2.14)	93.04(9.73)	95.81(0.21)	88.91(10.86)	97.18(0.40)	**97.60(0.76)**
6	10	266	76.79(11.48)	78.32(7.35)	70.34(7.73)	**97.38(2.56)**	82.61(6.23)	68.15(19.63)	66.08(11.56)	76.69(4.88)	88.52(7.86)	90.43(5.09)
7	10	714	95.94(3.64)	94.13(3.02)	91.84(4.42)	**95.75(0.35)**	94.60(6.83)	92.57(4.90)	94.50(7.53)	89.24(8.13)	94.23(4.75)	90.96(2.89)
8	10	736	91.80(11.16)	95.48(4.19)	92.61(5.52)	**98.81(0.83)**	97.14(2.10)	94.24(4.07)	89.81(5.36)	97.75(1.11)	97.85(0.94)	98.06(0.12)
9	10	69	46.06(19.36)	69.73(7.10)	63.15(5.98)	**93.42(8.30)**	82.32(13.90)	45.95(14.83)	57.66(15.13)	85.43(12.08)	91.76(7.18)	92.30(3.03)
OA (%)			89.06(8.00)	93.84(6.78)	90.12(12.02)	95.28(1.86)	93.68(0.96)	86.15(6.39)	85.45(9.22)	89.46(11.47)	94.92(1.66)	**95.46(0.56)**
AA (%)			81.95(7.59)	88.79(7.41)	83.47(10.36)	92.52(1.73)	89.77(2.17)	79.54(7.46)	79.25(8.17)	85.76(12.12)	92.28(1.91)	**93.15(0.49)**
Kappa (%)			87.01(6.72)	92.62(7.37)	88.22(9.98)	94.38(0.16)	92.46(1.04)	83.61(7.77)	82.74(11.50)	87.43(9.63)	93.93(0.78)	**94.57(1.05)**
Time (s)			95.83	143.38	105.24	119.35	145.87	248.82	250.48	359.45	396.54	414.69

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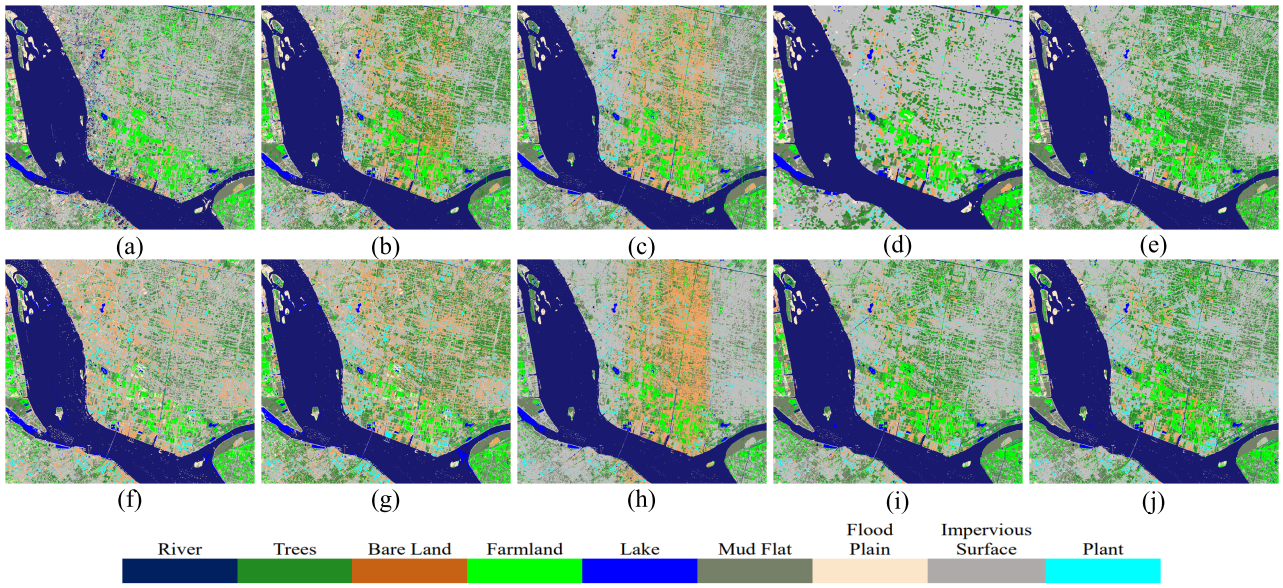


Fig. 10. Classification maps of Yangtze River Delta. (a) ELM. (b) 3-D CNN. (c) ProCRC. (d) KENLRR. (e) Kernel ProCRC. (f) RoF. (g) XGBoost. (h) TCRC-Bagging. (i) KProCRC-LNE. (j) KProCRC-LNSAE.

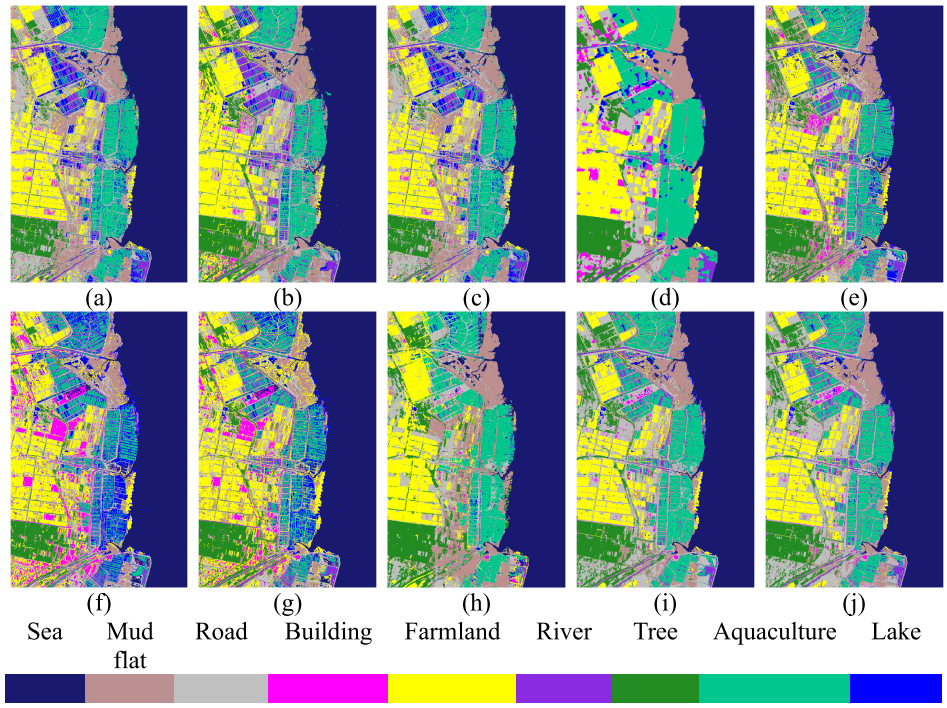


Fig. 11. Classification maps of Dafeng Natural Reserve. (a) ELM. (b) 3-D CNN. (c) ProCRC. (d) KENLRR. (e) Kernel ProCRC. (f) RoF. (g) XGBoost. (h) TCRC-Bagging. (i) KProCRC-LNE. (j) KProCRC-LNSAE. (k) Class legend.

achieve high accuracy but also clearly outline various ground objects and retain their detail texture. The RoF and XGBoost methods tend to confuse roads and buildings due to their similar spectrum. Since aquacultures are mostly in artificial water body, they are sometimes misclassified as lakes.

Through an overall analysis of the variance of classification performance in Tables V–VII, it is found that lower accuracy is often accompanied by larger variance. The reason may be that the results with low accuracy are greatly affected by the samples. Conversely, the results with higher accuracy are more robust. In general, the variances of ensemble learning method are smaller than those of single classifiers. The process of majority voting may reduce the impact of abnormal results. In comparison, the variances of the proposed methods are all at a low level, which indicates that the models are relatively stable and not easily affected by samples.

For the running time of these methods, it often takes hundreds of seconds to run due to the large size of data. Ensemble learning methods generally take a long time because they need to run more classifiers. Although the proposed Kernel ProCRC requires some time to calculate the Gram matrix in the kernel function, it can still complete the classification task efficiently.

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TABLE VIII
SIX CLASSES OF THE PURDUE CAMPUS DATASET

Class	Ground object	Number of samples
1	Road	1287
2	Grass	1114
3	Shadow	219
4	Soil	379
5	Tree	1351
6	Roof	1285
Total		5635

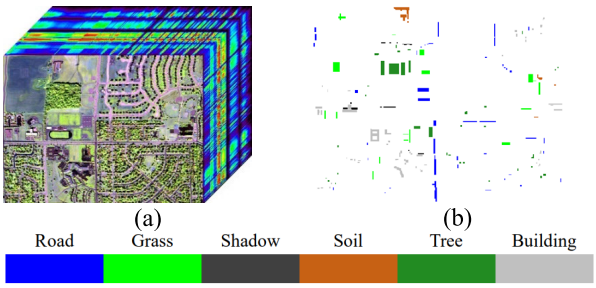


Fig. 12. (a) False-color image. (b) Ground truth.

V. DISCUSSION

A. Generalizability of the Methods

The three datasets in the experiments are all from GF-5. In order to verify the generalization performance of the proposed methods, relevant experiments are carried out on another classical HSI dataset. The dataset applied here was the Purdue campus, which was acquired by the hyperspectral mapper (HyMap) system. The image consists of 377 × 512 pixels with a spatial resolution of 3.5 m. These data cover 126 spectral bands. As shown in Table VIII, six classes of objects are contained, and the false-color composite image and ground reference image are shown in Fig. 12(a) and (b), respectively. The performances are listed in Table IX and the maps are shown in Fig. 13.

It is not difficult to see from the table that the two proposed ensemble learning methods rank the top two in OA (being 94.46% and 92.41%). XGBoost misclassifies part of the road as soil and TCRC-Bagging makes the misclassification of grass and soil. Although the OA of Kernel ProCRC is lower than that of KENLRR, our method retains almost all details, which facilitates subsequent image interpretation. The results indicate that the proposed methods are stable and adaptable even when applied to HSI data from different sensor sources.

B. Comparison of Bagging and Proposed Method

Sampling by bootstrap can obtain training sets with differences. Not all samples can be sampled in a subset so that the method can reduce the interference of abnormal samples. However, it might be a double-edged sword, especially for small size samples. It will result in some unrepresentative training subsets degrading the ability of classifiers.

To confirm this idea, four classifiers (SVM, naive Bayes, CRC, and KCRC) are used as basic classifiers to compare the performance of the bagging method and the proposed LNE method. The performance of the two methods (i.e., OA of ensemble result, each basic classifier, and average OA of all basic classifiers) is shown in Fig. 14.

From the point of view of basic classifiers, the OA of basic classifiers of the LNE is significantly higher than that of bagging. The mean accuracy of the basic classifiers of the LNE is also about 2% higher than that of bagging.

When it comes to ensemble results, LNE still outperforms bagging, while different basic classifiers perform differently under the frameworks of these two ensemble strategies: for SVM, LNE is 3.6% higher; for naive Bayes, LNE is 0.4% higher; and for both CRC and KCRC, it is about 1% higher.

C. Analysis of Parameter Sensitivity

For Kernel ProCRC, parameters λ, γ, and σ significantly impact the performance. λ and γ are related to the ProCRC, adjusting the weights of three terms in (7). The first is the reconstruction term, which makes the combination of atoms consistent with querying samples. The second is the collaborative term, which avoids the appearance of large coefficients and encourages all atoms to represent the querying samples. The third is the adjustment term, which makes the combination of the intraclass atoms similar to the querying sample.

Some empirical parameters were found. When $\lambda = 1e-3$, $\gamma = 1e-5$, and $\sigma = 0.1$, the experiments on three HSIs obtained almost the best classification results. Therefore, when the parameters λ and γ were adjusted, σ was fixed to 0.1; when adjusting σ, set $\lambda = 1e-3$ and $\gamma = 1e-5$.

The grid search method is used here to explore the effect of each parameter on the classification and find the optimal parameters. Specifically, λ and γ are selected from [1e − 5, 5e − 5, 1e − 4, 5e − 4, 1e − 3, 5e − 3, 1e − 2] and the range of σ is set from 0.02 to 0.5 in steps of 0.02.

For three different datasets, the relationships between λ, γ, and OA are shown in Fig. 15. It can be seen that, when the value of λ is greater than 1e − 3, the change of γ does not have a great impact on OA, and OA is maintained at a high value. At this point, the collaborative term plays a leading role. When the value of λ is less than 1e − 4, the OA is greatly influenced by γ. As γ increases, OA increases slowly at the beginning to the top, then decreased slowly, and finally decreased sharply to a very low level. At this time, the role of collaborative term is limited, and the adjustment term plays a leading role. As λ has taken an appropriate value, γ becomes the main factor that affects OA. When γ is very small, this term has little effect. The classifier is the same as Kernel CRC. However, while γ is very large, the strong influence of the adjustment term makes the first two terms ineffective, which leads to a drop in the ability of the classifier.

Besides, σ is related to the Gaussian function. From Fig. 16, it can be easily found that the curve generally shows a trend of rapid growth at the beginning and then a slow decline, but for different data, the peaks appear at different values of σ. For the Yellow River Delta and Yangtze River Delta data, when σ is nearby 0.1, the classification performance is the best. For Dafeng Natural Reserve data, the OA keeps at a high level when σ is between 0.1 and 0.2.

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TABLE IX
CLASSIFICATION ACCURACY (%) FOR THE PURDUE CAMPUS DATASET

Class	Train	Validation	ELM	3D CNN	ProCRC	KRNLRR	Kernel ProCRC	RoF	XGboost	TCRC-Bagging	KProCRC-LNE	KProCRC-LNSAE
1	10	1277	79.79(11.92)	92.06(9.99)	83.61(6.39)	**92.18(7.41)**	84.23(9.84)	77.63(14.31)	77.09(7.30)	74.69(13.84)	91.21(8.26)	88.80(5.66)
2	10	1104	88.74(5.26)	93.06(6.34)	93.63(4.22)	91.48(5.93)	95.98(1.21)	92.58(3.76)	93.05(5.81)	85.32(6.57)	94.27(3.19)	**97.90(0.10)**
3	10	209	55.48(17.22)	84.57(7.40)	**91.75(3.12)**	65.89(9.53)	77.19(7.52)	66.17(8.35)	67.22(8.72)	88.89(6.03)	81.49(9.13)	84.95(3.02)
4	10	369	90.12(2.96)	80.27(9.64)	92.51(7.02)	91.60(6.34)	83.24(10.89)	72.66(8.12)	71.97(13.02)	57.05(10.52)	80.35(8.84)	**94.68(8.25)**
5	10	1341	96.01(4.01)	99.31(0.65)	93.94(6.27)	97.77(0.33)	98.89(0.56)	97.08(1.92)	96.88(5.99)	92.09(6.44)	**99.13(0.15)**	98.66(0.61)
6	10	1275	90.28(7.12)	89.77(11.30)	90.44(8.96)	**99.00(0.10)**	94.99(7.75)	86.00(16.66)	86.26(18.61)	90.34(9.15)	94.25(5.62)	96.81(2.79)
OA (%)			86.05(9.31)	92.15(7.63)	90.02(5.90)	92.41(4.01)	90.84(4.80)	84.70(9.32)	84.64(6.02)	81.23(6.59)	92.41(3.51)	**94.46(2.53)**
AA (%)			83.40(10.69)	89.84(8.85)	90.98(6.86)	89.65(6.58)	89.08(4.42)	82.02(8.60)	82.08(6.83)	81.40(5.50)	90.12(2.64)	**93.63(1.34)**
Kappa (%)			82.52(9.55)	90.12(8.68)	87.36(9.51)	90.52(6.63)	88.52(4.99)	80.89(7.75)	80.82(7.20)	76.36(7.01)	90.48(3.32)	**93.03(1.48)**
Time (s)			34.26	63.55	32.72	46.63	56.21	76.64	104.26	238.24	297.95	345.67

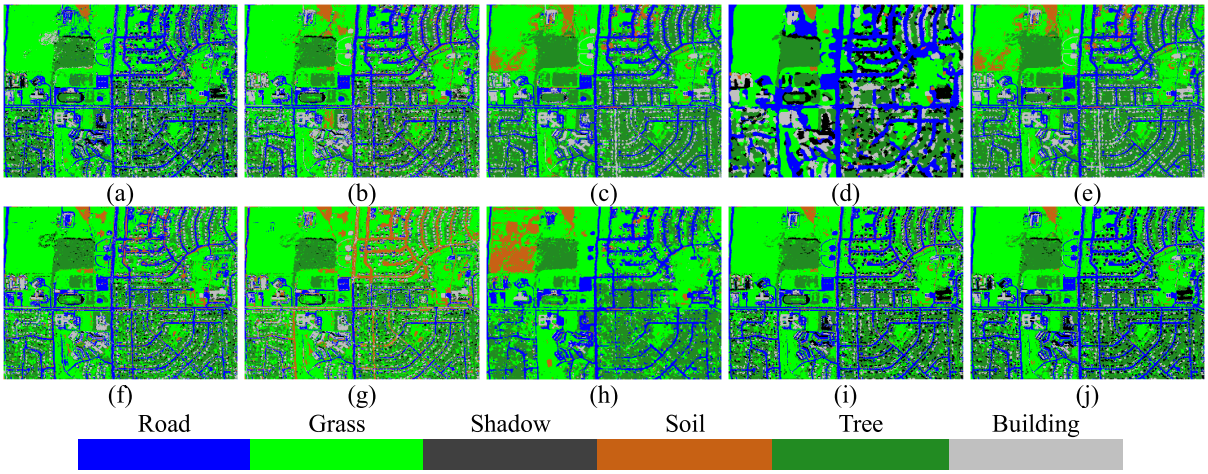


Fig. 13. Classification maps of Purdue Campus. (a) ELM. (b) 3-D CNN. (c) ProCRC. (d) KENLRR. (e) Kernel ProCRC. (f) RoF. (g) XGBoost. (h) TCRC-Bagging. (i) KProCRC-LNE. (j) KProCRC-LNSAE.

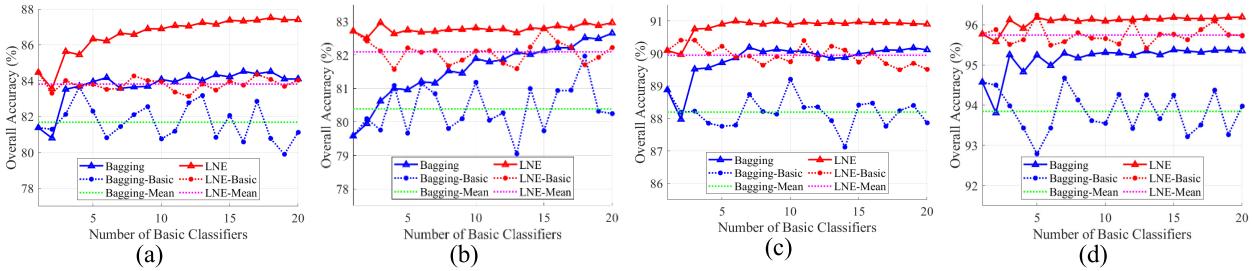


Fig. 14. Performance of bagging and LNE. (a) SVM. (b) Naive Bayes. (c) CRC. (d) KCRC.

The size of the neighborhood, adjusting by R, determines the number and location of candidate samples. If R is small, there are a few samples to choose from, leading to the selected samples that have high similarity due to their close geospatial distance. If R is large, the probability of selecting the wrong sample also increases.

The multiple of the number M determines the number of atoms in the dictionary. When $M \geq 2$, the number of atoms increases to achieve the purpose of sample augmentation.

A series of experiments is conducted to explore the influence of R and M. The results are shown in Fig. 17. They record the OA under different M and R values. The solid line represents the OA of the ensemble result, while the dashed line represents the mean value of OA of basic classifiers.

For KProCRC-LNE, the relationships between OA and R are shown by green solid and dashed lines in Fig. 17. As R increases, the OA of basic classifiers gradually decreases, while that of the ensemble results first increases and then decreases. It is easy to interpret based on the premise that samples farther from central samples are of poorer quality. As the neighborhood expands, the diversity of samples increases, while the quality decreases. For the ensemble results, the increase in OA is due to diversity. When the neighborhood is large enough, the effect of diversity is limited. Then, the degradation of sample quality plays a dominant role.

For KProCRC-LNSAE, the influence of M on OA is analyzed first. Here, M is set from 1 to 4 for comparison and analysis. It is noteworthy that as M increases from 1 to 3, the OA is obviously improved. However, when $M = 4$, the performance is similar to that when $M = 3$. Then, the influence of R on OA when $M = 3$ is analyzed. Like KProCRC-LNE,

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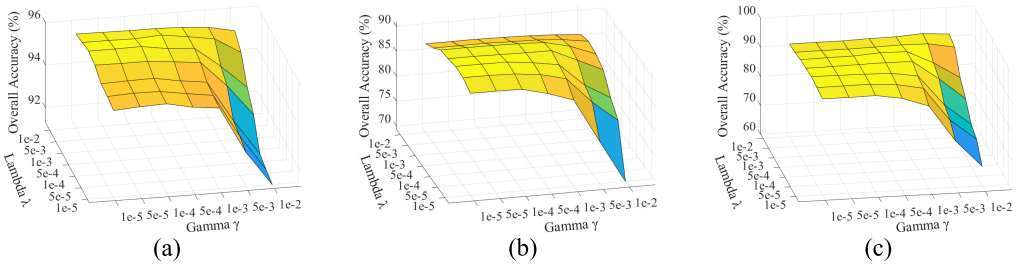


Fig. 15. Classification OA versus λ and γ. (a) Yellow River Delta. (b) Yangtze River Delta. (c) Dafeng Natural Reserve.

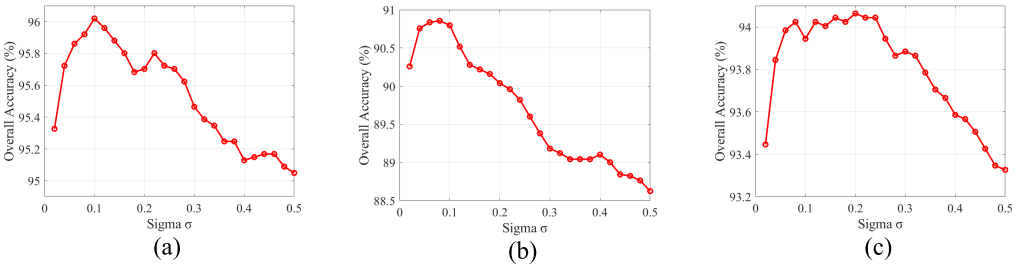


Fig. 16. Classification OA versus σ. (a) Yellow River Delta. (b) Yangtze River Delta. (c) Dafeng Natural Reserve.

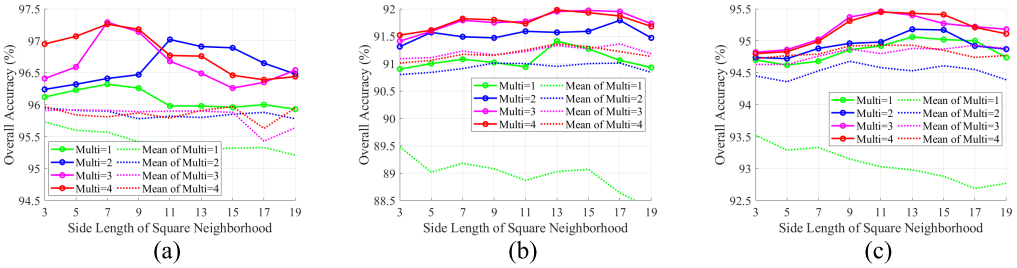


Fig. 17. Classification OA versus R. (a) Yellow River Delta. (b) Yangtze River Delta. (c) Dafeng Natural Reserve.

with the gradual increase of the parameter R, the OA also shows a trend of increasing first and then decreasing.

It is worth noting that even if the size of the dictionary changes when M changes, the optimal values of λ, γ, and σ are almost unchanged. This is of great significance in practical applications: as long as the optimal parameters of Kernel ProCRC are known, no matter how M and R in KProCRC-LNE and KProCRC-LNE change, the almost best results can be obtained without adjusting λ, γ, and σ.

D. Ablation Study

The proposed framework mainly consists of kernel methods, neighborhood information, and sample augmentation for classification. To assess the contribution of key components of KProCRC-LNSAE, ablation analyses are conducted by removing each component from the entire framework.

There are three variants in the ablation analyses: 1) "ProCRC-LNSAE": the kernel method is removed; 2) "KProCRC-Bagging": the neighborhood information is deleted; and 3) "KProCRC-LNE": the sample augmentation is deleted form KProCRC-LNSAE. The mean accuracy and Kappa of ten experiments for these ablation models are listed in Table X. It is obviously shown that the whole KProCRC-LNSAE outperforms other variants and gains great improvements. The OA drops significantly from 2% to 5% in ProCRC-LNSAE, where it is difficult to achieve the expected performance without kernel methods. Compared with bagging, the neighbors play key roles in LNSAE resulting in the improvement of OA. There is no sample augmentation in KProCRC-LNE, and the performance drops lightly due to the lack of quality samples.

TABLE X
ABLATION COMPARISON OF EACH VARIANT OF KPROCRC-LNSAE

Model	ProCRC-LNSAE	KProCRC-Bagging	KProCRC-LNE	KProCRC-LNSAE
Data set	Yellow River Delta			
OA (%)	92.28	95.88	96.32	**97.29**
KC (κ)	90.16	95.53	96.00	**97.05**
	Yangtze River Delta			
OA (%)	87.27	90.01	90.15	**91.84**
KC (κ)	86.36	87.99	88.28	**90.26**
	Dafeng Natural Reserve			
OA (%)	91.06	93.96	94.92	**95.46**
KC (κ)	90.23	92.03	93.93	**94.57**

E. Performance on Small Size Samples Situation

The performance of the classification methods in the case of small size samples is an important criterion. To test the performance of these methods with few samples, all the comparison methods are divided into two groups: the single classifiers and the ensemble learning methods, and compared the effect of different sample sizes on OA. Here, the number

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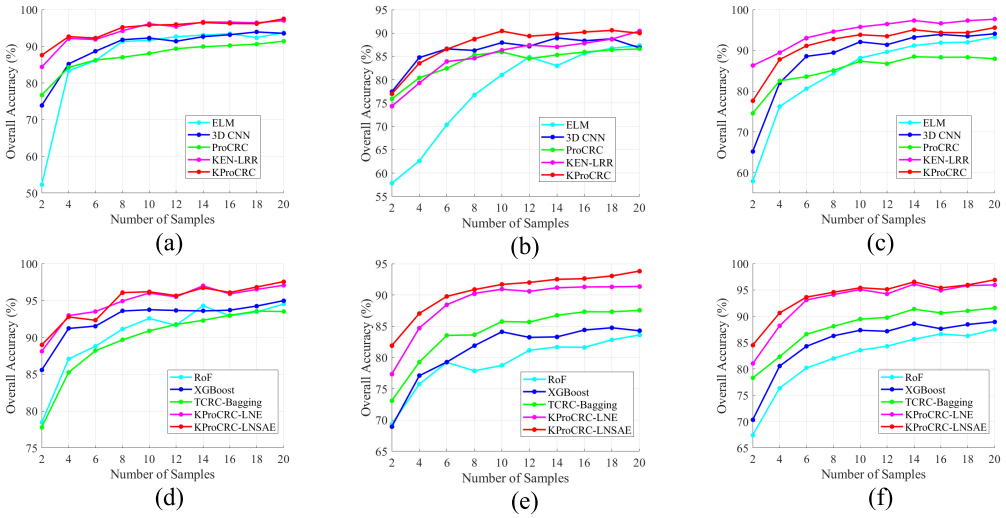


Fig. 18. Classification OA versus number of samples per class. (a)–(c) Single classifiers for Yellow River Delta, Yangtze River Delta, and Dafeng Natural Reserve, respectively. (d)–(f) Ensemble methods for the tree data.

of samples for each class is [2, 4, 6, . . . , 18, 20]. The results of experiments are shown in Fig. 18.

For single classifiers, their OA difference with few samples is greater than that with many samples. Specifically, for Yellow River Delta data, when the sample size is only 2 per class, the OAs of ELM, 3-D CNN, and ProCRC are less than 80%, while those of KENLRR and Kernel ProCRC achieve 85.6% and 88.3%, respectively. For Yangtze River Delta data, the OA of ELM is much lower than that of other methods when samples are few. For Dafeng Natural Reserve data, 3-D CNN achieves good performance with many known samples, but it does not perform well with few samples.

For ensemble learning methods, on Yellow River Delta data, with few samples, the two methods based on Kernel ProCRC outperform other methods. When the number of training samples is only 2 per class, their OA can be more than 85%. OA of all methods can achieve over 90% when the number of samples exceeds ten. For Yangtze River Delta and Dafeng Natural Reserve data, the two proposed methods achieve excellent performance. Among the proposed methods, KProCRC-LNSAE gains the best performance due to sample augmentation.

VI. Conclusion

In order to better solve the problems of HSI classification of wetland, a framework is proposed in this article. Kernel ProCRC can solve the problem of linear inseparability well. The LNE and LNSAE methods make full use of the neighborhood and label information of the known samples. This framework can accurately classify wetland species.

In this article, three datasets of coastal wetlands are used for experiments. The results show that the proposed methods achieve excellent performance compared with other state-of-the-art methods. The performances of the mentioned methods in the case of small size samples are also explored with satisfactory results. The accurate classification maps with detailed texture information generated by our methods can provide the necessary conditions for postprocessing and application of wetland data.

There are still some shortcomings in the proposed methods. Only the Gaussian function is used in the kernel method, which causes some limitations when facing different datasets. Also, a random method is adopted when selecting samples in ensemble learning. It undoubtedly increases the instability of these methods. Problems can be solved from the following aspects. Multiple kernel learning can combine different kernel functions. In this way, the difficulty of selecting kernel functions and the performance limitation of single kernel function can be avoided. To make basic classifiers perform better and reduce the randomness of the classification results, the difference and similarity of neighborhood samples can be considered when selecting them. These are what we will research in the future.

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